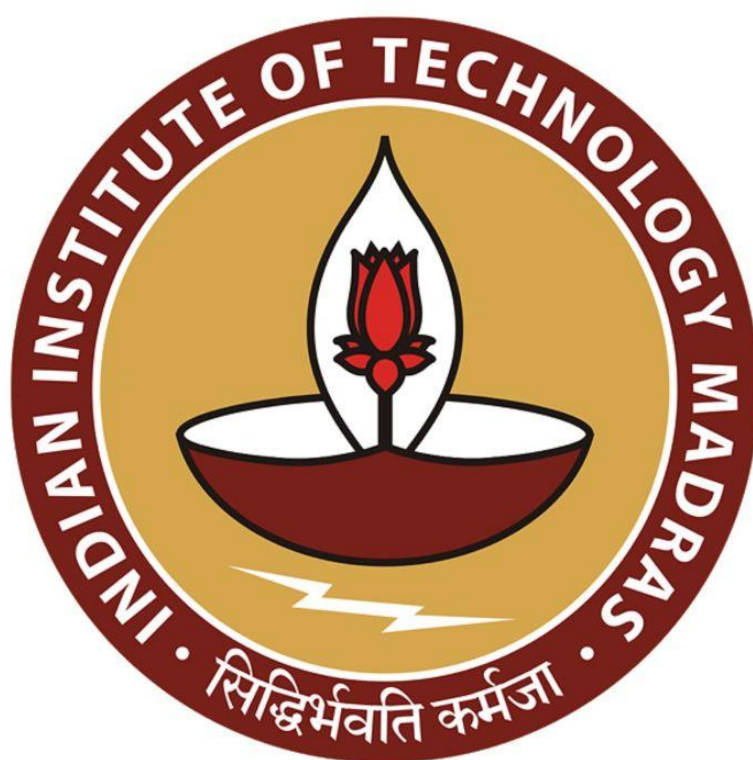




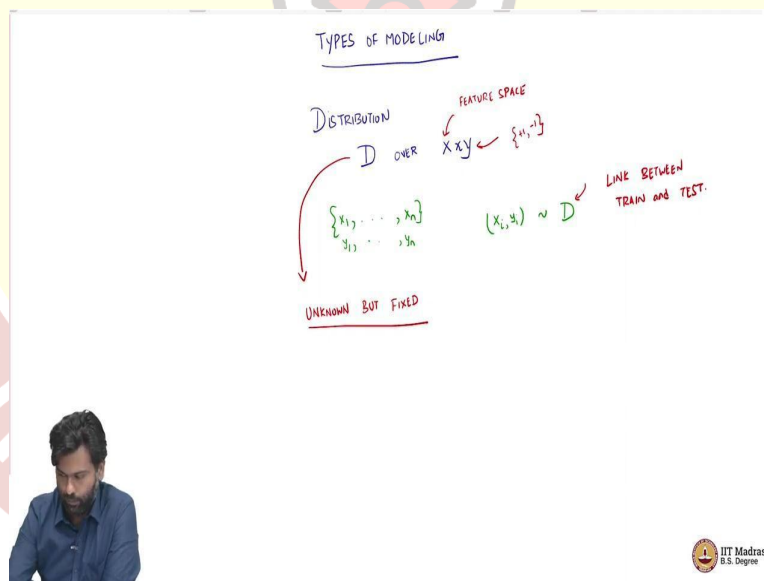
IIT Madras
ONLINE DEGREE

Machine Learning Techniques
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Generative and Discriminative Models

Hello everyone, welcome back. So, far in this course, we have been looking at supervised learning problem. And we have looked at two major algorithms for the binary classification problems specifically under supervised learning. And those were the K nearest neighbor algorithm and the decision tree algorithm. So, both of these are, in some sense, intuitively simple algorithms that you can develop, based on your intuition as to how you want your model to be in the case of decision tree, or in general, what is the first thing that comes to your mind when you want to solve the classification problem in the case of K nearest neighbors.

What we are going to do today is look at something very important, which is a broad perspective of how to do modeling in machine learning itself.

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So, today's goal is to understand types of modeling. So, what do I mean by this? The first thing we ask is, in machine learning is that you have some training data, and then you build some model based on this training data. But then you are going to be tested on the test data, which is not necessarily the same points as that you have seen in the training set. So, which means that what we have in training data, I mean, of course, we know that if you just

memorize, you are not going to do perhaps on the test set. This we have already kind of discussed at a high level.

Now then, if the test data is completely irrelevant to the training data. For example, you study for a physics test, but then you are tested on chemistry. Of course, you are not going to be able to do well on that test, because what you are being trained on is on a different set, but then you are being tested on something else. So, that is also hard. Like, that is an impossible task. So, the question then we ask is if we allow test data to have exactly the same points as training data, then memorization would work, and that is not going to be so useful in practice.

On the other hand, if we make test data completely irrelevant to the training data, then no algorithm can do any learning. So, because learning itself is not possible, because what you can learn is only from training data, but then if the test data is something else, then of course, we cannot do anything useful. So, now, we need to come up with a way to connect the train and the test data. So, unless there is some information that we can gain from the train data, which is of use to us, when we are actually doing the test data, when we encounter the test data and make our predictions in the test data, unless there is some relation, there is no hope of performing or there is no hope of meaningful learning from the training data.

So, the question is, how can we assume that this relation exists, what kind of assumptions we should make, to formalize this kind of a relationship. And the main assumption that one would make in a machine learning setup, it is a very common assumption, is that there is some distribution. Let us call this d over x cross y . What is x ? x is our feature space. You can think of this as, let us say, the D dimensional space where you have D features, your y in a classification setup could be $+1$ or -1 .

Of course, in a regression setup, this would be reals. So, now, what is the assumption here? This is one way to assume that there is some distribution over the feature cross labels. What does this mean? This means that every data point that you are seeing in your training set is drawn according to this distribution. So, we have our training set, which has $x_1, \dots, x_n$, and $y_1, \dots, y_n$. We will talk about classification for now but then whatever we are discussing also applies to other supervised learning setup. So, now what is it that I am going to assume?

I am going to assume that you know every pair $x_i y_i$, that is the feature and the corresponding label is drawn according to this distribution D . So, think of this distribution as some kind of a probabilistic mechanism that generates our data. So, if you remember our estimation

discussion, we had this box with some parameters inside and then every time you press a button a data point came out. There it was only the features, now, we are saying that you have a data point and the corresponding label also come out. So, this box could be as complicated as possible, may not be just simple coin tosses or anything of that sort, might be a complicated box, but then if I press that, once I get a single data point x_1, y_1 , we press it again I get x_2, y_2 and so on, I get x_n, y_n .

So, this is how my training data set is generated. There is some underlying probabilistic mechanism that is generating this. Now, what about the connection between training and test set? The connection is via this underlying distribution D . So, we are saying that the test set is also going to come from the same underlying distribution. So, now, you are going to also draw a data point from the same distribution D and then when you are trying to do the testing, you are going to be tested on some x test and y test, but then you are only shown the x test, and you need to predict what the y test is. So, then the link between the training data and the test data is the fact that we assume there is some underlying distribution D .

So, this is the link between train and test. So, which means if we understand the characteristics of this distribution then we kind of, the hope is that we will be able to use that to make predictions. So, that is the hope. At this moment, we are not making absolutely any further assumptions about this distribution D , we are not saying any specific distributions associated, for example, we are not saying that the features are Gaussian distributed, we are not saying the labels are Bernoulli distributed, nothing of that sort at this point.

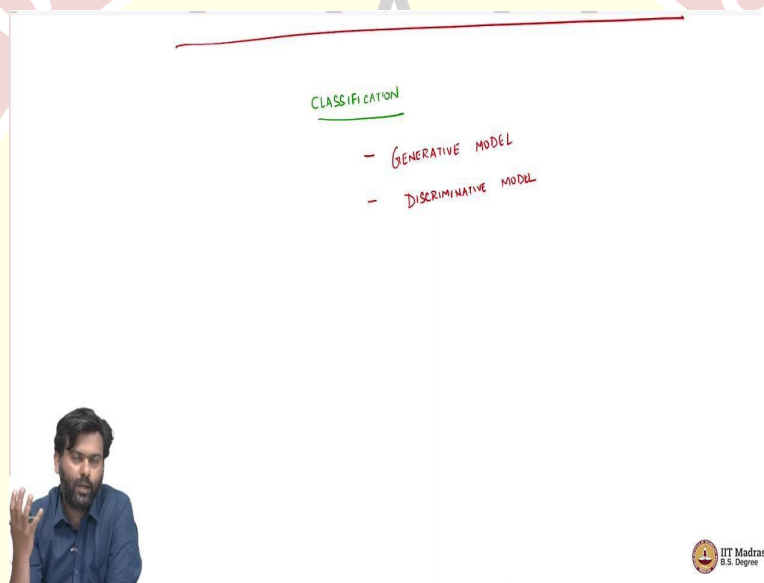
So, at this point, we are saying there is some D and the only thing we are assuming about this D is unknown but fixed. So, we do not know, as an algorithm, we do not know what this distribution D is, the only thing that you have, which gives you information about this distribution, D is your training data and you need to learn your model from this training data and hope that if your learning is good, then the model that you have gotten from the training data, captures succinctly what the distribution D is, so that you can use this model to make good predictions on the test data.

At a high level this is the idea. So, this is a high level paradigm. So, now, what we are going to do now is then, go one step further and say that to develop a model, we need to make some assumptions on the underlying distribution D . Now, depending on what assumptions we make on the underlying distribution D we are going to get different algorithms. So, every algorithm

is going to somehow cater to this underlying distribution, D 's assumption that we are making, and that will lead us to different sorts of algorithms.

So, now what kind of assumptions we can make on this underlying distribution D ? We know it is unknown, we know it is fixed. So, this is a common assumption, assumption number 1, which we are making. In addition to this, can we make more some more structural assumptions on the distribution D itself? So, that is where there are two major schools of thoughts, which lead us to two different types of modeling itself. So, what are these two types of modeling? Let us see that now.

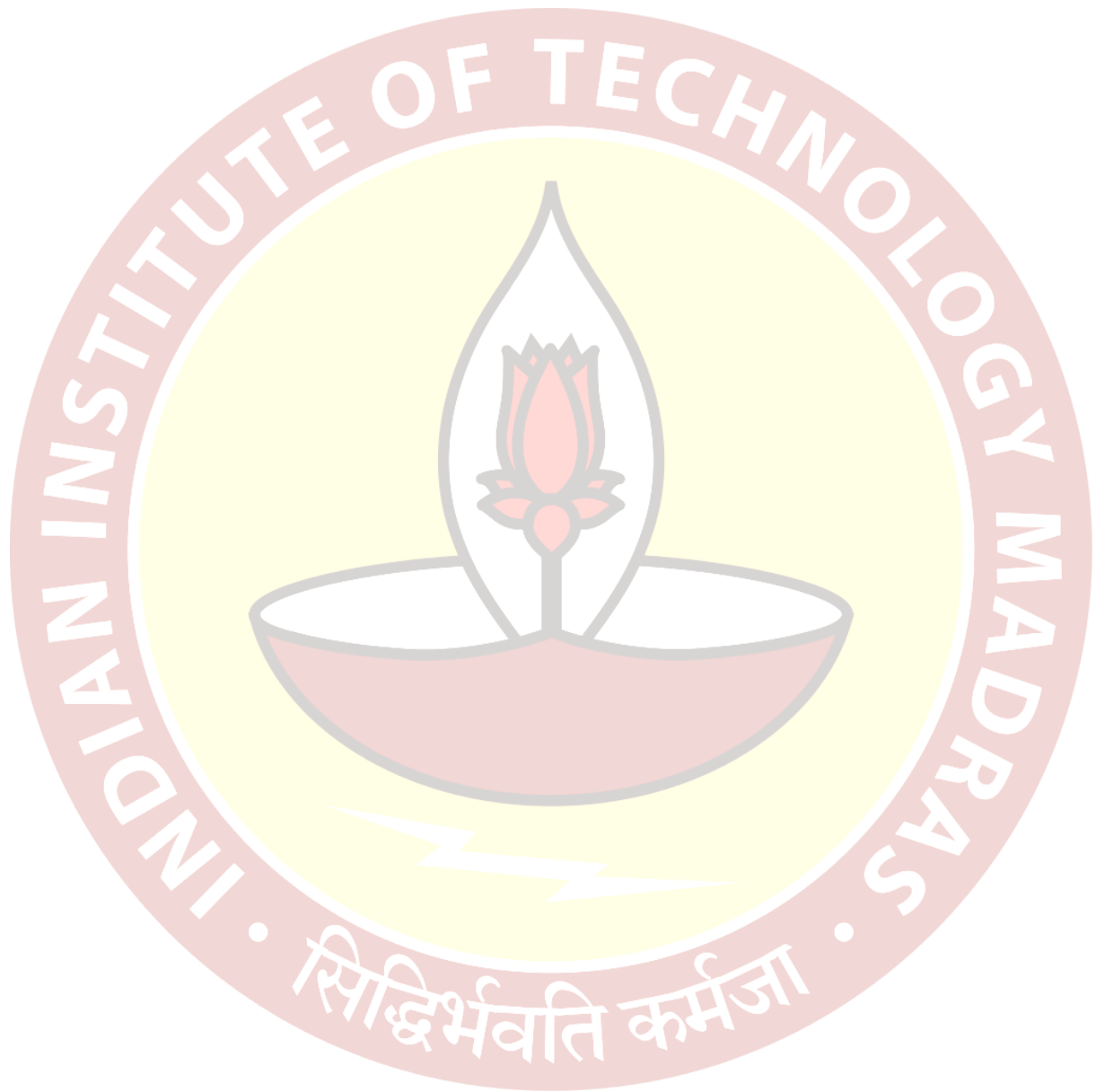
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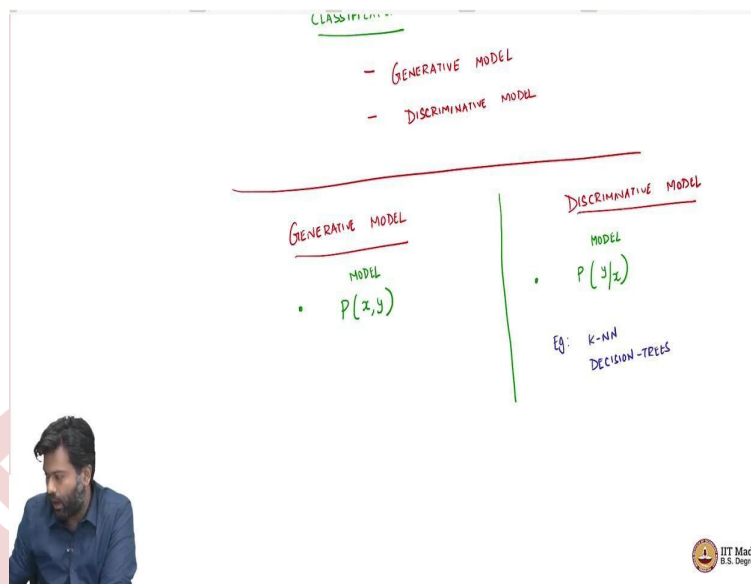
So, we are still in the classification setup. We can think of similar things for the regression also, but then I will talk about the classification examples. So, the two major schools of thoughts with when it comes to modeling are the following. One is what is called as a generative model and we will talk about what it means in a minute and the other is what is called as a discriminative model. Remember, we are somehow trying to make further assumptions on the underlying distribution D over the feature, label space that generates our training and test data.

So, there are two different ways to do this modeling. One modeling is the generative model. The other modeling is what is called as the discriminative model. I will first tell you what these models are, what is the subtle but very key difference between these two models are and then we will talk about what does it mean in practice to say you are doing a generative

model or a discriminative model, and that will give us some insights into why people have two different types of modeling.



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So, here is the key difference between generative model and discriminative model. So, you have a distribution over X and Y . So what does that even mean? That means that there is some probability or a density associated with every feature, and label, so there is some $P(x,y)$. That is, if I tells you hey, this is my feature vector, and this is the corresponding label, then we should be able to associate it with the density or a probability. Density, when your features are continuous; probability, when your features are discrete.

Why? Because we have discussed that. So, when you have continuous random variables, then the probabilities for any feature vector will become 0, that is useless. So, we will model the density. So, using a density. So, this is a generative model. So if you are putting, if you are making assumptions on $P(x,y)$, how should this density look like? Then it means that we are doing a generative modeling. I will talk about why it is called generative modeling in a bit, once I described what is the discriminative model.

The discriminative model, on the other hand, models, not $P(x,y)$, that means it does not model the joint distribution of features and labels. But it only models $P(y|x)$. So, this guy models $P(x,y)$ discriminative models, $P(y|x)$. So, this is the fundamental difference. So either modeling the joint distribution of x , y , or just modeling $P(y|x)$. Now what does it even mean to do say that there are two different types of modeling?

To understand this better, let us take a simple example. I am going to give this example from my point of view. But from your point of view, you can think of an analogous example very easily once you see what the example is. Let us say I am given the task to do the following. It

is a binary classification task where I will be given some spoken sentence and I need to say whether the sentence is in Bengali or not. This is the task that I am given.

I am taking Bengali as an example, because I do not know how to speak or write Bengali. That is why I am taking this as an example. So, it is a binary classification example where I am given this task where, by looking at the audio signal of some sentence, I need to classify it as whether it is a Bengali sentence or a non-Bengali sentence. Now, of course, I have a lot of training data and from which I need to train a model. So, now I need to make some assumptions about how this data could have been generated.

So now, here is where the generative model and the discriminative model differs. The generative modelers who tell you that model, how Bengali sentences themselves are generated. So, in the sense that you are modeling x , x here is a feature vector, which is this Bengali sentence that is being given to me, if you are modeling how the features themselves are generated, and how the labels are generated, given the features. So, you are doing both of these, you are doing a joint modeling of both features and the labels, then it is called generative modeling.

It means that I am making some assumptions about how Bengali sentences are generated. So, that is the key assumption in a generative model, where you are also modeling the feature generation process. On the other hand, the discriminative modelers who tell me, your goal is only to make sure whether the given sentence is Bengali or not Bengali. That is, your goal is to discriminate whether a given sentence is of a particular class or of a different class. In this particular case, it is a Bengali class or a non-Bengali class.

To learn, to be able to acquire this ability to discriminate between Bengali or non-Bengali sentences, you do not necessarily need to know how the Bengali language sentences themselves are generated. So, you do not need to know how to write or read Bengali to generate Bengali sentences, to just discriminate whether a sentence is in Bengali or not. That is, if you only want to say, what is the y label, given a feature x , you only want to model how is y given x looking like. So we only want to model this in a probabilistic fashion, we do not have to care about how the x itself is generated.

The generative modelers would say no, no, no, you should also care about how the x is generated, how the features are generated. So, this is the subtle difference. If you are also modeling the feature generation somehow, then it is a generative model; if you are only

caring about how to discriminate whether a given feature is in label y equal to 1 or y equal to -1, then that is a discriminative model. Now, one might ask, which is a better model?

So now, I, as a person who does not know how to speak Bengali, will still be able to do reasonably well given a sentence to say whether it is Bengali or not, but if you asked me to speak in Bengali, I will not be able to do at all because I am not trained in Bengali. Now, let us ask another person who actually can speak Bengali and give that person the same task. Now that person is going to ace this task, because that person has the ability to generate Bengali sentences. So, this task is going to be very, very easy for that person as compared to me.

Now, what does this mean? This tells us that now that person who can speak Bengali has a mental model of how to generate Bengali sentences, so that person knows the generative process itself. And so it becomes easy for that person to do a discriminative task. The task is at the end of the day to discriminate given a sentence Bengali or not. So, if you know how to generate then you are able to discriminate well. So, on the other hand, if I try to model using my, if I try to force a mental model of the generative process of Bengali into my mind, now, that is not going to work because I do not know how to speak or write Bengali.

So, the high level takeaway then is the following. If you have reasons to believe how the exact features are generated in your task at hand, then it is a good idea to do generative modeling. Because if you know how to generate correctly, then you are usually better off in doing a discriminative task better as well. On the other hand, if you have no idea how the features are generated, then it may not be a good idea to do a generative modeling because if your model is wrong, then you would be suffering even in the discriminative task as well.

So, if your domain expertise is such that you know exactly how the data could have been generated, the features could have been generated, then generative modeling is a good idea. Otherwise, discriminative modeling is a good idea. So, at a high level, this is the takeaway that I want you to think about for now. Now, what we are going to do is we are going to look at a lot of examples of algorithms, which makes specific generative assumptions and specific discriminative assumptions and then we will be able to appreciate this even better. So, when should you do a generative assumption versus when you should do discriminative assumptions?

There is no single answer to this because both are typically useful, but the broad idea is that if you do not have much idea about how the features are generated, then it is better to stick to a discriminative modeling based assumption. So, which is what a lot of algorithms that we will see, this course will also do. In fact, that brings us to this point that we have already kind of seen two algorithms. Now we can ask the question whether these are generative model based assumptions or discriminative model based assumptions. what are the two algorithms we saw?

One is k nearest neighbor, the other is decision tree. Now in both these algorithms, we have a model in decision tree, you have a model; in nearest neighbors, the model is the entire data set. Nevertheless, if I want to call that a model, in both these algorithms, the commonality is that, I am not bothered about how the data features themselves was generated. The only thing I am bothered about is if a new feature is given, or if a feature is given, how do I model what the label for this is. In the nearest neighbor case, I would say that, for every data point, the label is based on the K nearest neighbors.

So, I do not care about how the x itself is generated. I am only specifying how to make a prediction of y if x is given. So, essentially we are doing discriminative modeling and nearest neighbor. The same thing in decision tree as well. I do not model how x is generated, I do not have any distributional assumption or any assumption as to how the data features are generated. But I make the assumption that, if a feature is given, then the decision tree would be, traversing through the decision tree would give me the label.

So, I am able to discriminate whether a given feature is y equals 1 or y equals -1, using the decision tree that I have built from data. So, both these that we have seen so far, our K nearest neighbors and decision tree are examples of discriminative models.

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The whiteboard is divided into two sections. The left section is titled 'GENERATIVE MODEL' and contains the text 'MODEL' and the equation $P(x, y)$. Below this, it says 'NEXT'. The right section is titled 'MODEL' and contains the equation $P(y|x)$. Below this, it says 'Eq: K-NN' and 'DECISION-TREES'. It then shows the equation $P(y=1|x) = 1$ if 'decision tree for x says 1.' and $P(y=1|x) = 1$ if 'majority of neighbours say 1' and $= 0$ otherwise.

GENERATIVE MODEL

MODEL

$P(x, y)$

NEXT

MODEL

$P(y|x)$

Eq: K-NN

DECISION-TREES

$P(y=1|x) = 1$ if decision tree for x says 1.

$P(y=1|x) = 1$ if majority of neighbours say 1

$= 0$ otherwise

But then one asks the question, what is the probability in this? Where is probability coming into the picture? Because if I am going to say $P(y|x)$, I need to make some probabilistic assumption. So, what is the probability here? This is not really a probability, but then you can think of it as a trivial probability. So, in the sense that what would the probability corresponding to this entry would be? So, it will say probability of y equals 1 given some x , so the class that the given test given point x belongs to this class 1, this probability is 1 if you know the decision tree for x says 1.

Otherwise, this probability is 0 which means the decision tree would have said 0 or -1 depending on what your label is. So, this is a trivial probability. So, the probability can be either 1 or 0, which means that there is a deterministic decision that we are making for every data point. So, that is why I mean we can think of always probability as like a generalization of your deterministic decision making. Similarly, in K nearest neighbors, you would say that probability of y equals 1 given x is 1 if majority of neighbors say 1 and the probability is 0 otherwise. Same thing is here, probability is 0 if that condition does not hold.

So, these are deterministic mappings from input space to the output space. Input space is the features, output space is +1 or -1 or +1 or 0. So, all deterministic mappings fall under this category of $P(y|x)$. So, because you are mapping given an x , you are just saying what the y is. So, we have seen already two examples, implicitly they are trivially discriminative models.

What we are going to do next is look at an example for a generative model. So, we will look at one algorithm where which makes a generative modeling assumption and then it will try to

learn some nice interesting models once this assumption is put down carefully. So, the next thing that we are going to see is this. So, this is what we will see next, an example for a generative model based algorithm.

